Exercise Sheet 6

Social Context

5942UE Network Science

26 March 2026

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6.A Homophily and Measurement

Exercise 6.A.1: Measuring Homophily by Hand

A friendship network of 8 students: 4 study CS (C_1–C_4) and 4 study Business (B_1–B_4).

Edges: C_1-C_2, C_1-C_3, C_2-C_3, C_3-C_4, C_4-B_1, B_1-B_2, B_2-B_3, B_3-B_4.

1. Count edges within CS, within Business, and across departments.
2. Compute the **E-I index**: $(E - I)/(E + I)$, ranging from -1 (pure homophily) to $+1$ (pure heterophily).
3. Is this network homophilic, heterophilic, or neutral with respect to department?
4. What would the E-I index be if ties formed **randomly** regardless of department?

Solution:

1. Edge counts: Within CS: 4 ($C_1C_2, C_1C_3, C_2C_3, C_3C_4$). Within Business: 3 (B_1B_2, B_2B_3, B_3B_4). Across: 1 (C_4B_1). Total: 8.

2. E-I index: $E = 1, I = 7$. $\text{E-I} = (1 - 7)/(1 + 7) = -0.75$.

3. Interpretation: Strongly negative → strong **homophily**. Students overwhelmingly befriend others in the same department.

4. Random baseline: With two groups of 4, expected cross-group fraction is $1 - (\binom{4}{2} + \binom{4}{2})/\binom{8}{2} = 1 - 12/28 \approx 0.57$. So 57% of edges should be cross-department under randomness; here only 12.5% are. The E-I index under random mixing would be $\approx +0.14$.

Exercise 6.A.2: Homophily vs. Confounding

Using `nx.karate_club_graph()` with the `"club"` attribute as the group label:

1. Count within-faction and cross-faction edges.
2. Compute the **E-I index** for faction.
3. Compute the expected cross-faction edge fraction under **random mixing**.
4. Is the network homophilic? By how much relative to random?

Solution:

```
from math import comb
import networkx as nx

G = nx.karate_club_graph()
factions = nx.get_node_attributes(G, "club")

internal, external = 0, 0
for u, v in G.edges():
    if factions[u] == factions[v]:
        internal += 1
    else:
        external += 1

ei_index = (external - internal) / (external + internal)
observed_cross_fraction = external / G.number_of_edges()

group_sizes = [
    sum(1 for faction in factions.values() if faction == label)
```

The karate club has 67 within-faction edges and 11 cross-faction edges, so the E-I index is

$$\frac{11 - 67}{78} \approx -0.718.$$

The observed cross-faction fraction is $11/78 \approx 0.141$, far below the random-mixing baseline of ≈ 0.515. The network is therefore strongly homophilic with respect to the final faction split.

6.B Affiliation and Similarity

Exercise 6.B.1: Three Closure Types

Affiliation network: students S_1, S_2, S_3, S_4 and courses C_1, C_2, C_3.

Enrolments: S_1–C_1, S_1–C_2, S_2–C_1, S_3–C_2, S_3–C_3, S_4–C_3.

1. Project onto the student layer (two students connected if they share a course).
2. Identify one example each of: (a) **triadic closure**, (b) **focal closure**, (c) **membership closure**.
3. If S_2 joins C_2, does this enable any new closure?
4. What information is **lost** when projecting?

Solution:

1. Projection: S_1–S_2 (via C_1), S_1–S_3 (via C_2), S_3–S_4 (via C_3).

2. Closures:

- (a) **Triadic:** (S_2, S_1, S_3) is an open triad — predicts S_2–S_3.
- (b) **Focal:** S_1 and S_3 both attend C_2 — the shared course is a focal context.
- (c) **Membership:** S_4 is friends with S_3 who attends C_2 — predicts S_4 may join C_2.

3. After S_2 joins C_2: S_2 now shares C_2 with S_1 and S_3, forming S_2–S_3 via focal closure. This simultaneously closes the triadic closure identified above — both mechanisms operate.

4. Information lost: (a) which course generated each tie, (b) the course nodes themselves, (c) tie-strength weights (multiple shared courses indicate stronger ties).

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Exercise 6.B.2: Bipartite Projection in Python

Implement the affiliation network from 6.B.1 in NetworkX and compute a weighted projection onto the student layer.

Solution:

```python
import networkx as nx

B = nx.Graph()
students = {"S1", "S2", "S3", "S4"}
courses  = {"C1", "C2", "C3"}
B.add_nodes_from(students, bipartite=0)
B.add_nodes_from(courses,  bipartite=1)
B.add_edges_from([("S1","C1"),("S1","C2"),("S2","C1"),
                  ("S3","C2"),("S3","C3"),("S4","C3")])

P = nx.bipartite.weighted_projected_graph(B, students)
# Weights = number of shared courses
```

The student projection reveals hidden social structure created by shared course memberships. Edge weights record how many shared affiliations exist between any pair.

6.C Social Dynamics and Causal Inference

Exercise 6.C.1: Distinguishing Social Forces

For each scenario, state whether it is best explained by **selection**, **socialization**, or **confounding**:

1. Two students both like jazz, become friends, and a year later both discover classical music.
2. Two colleagues at a tech startup both use Python; their friendship forms day 1.
3. Students in the same dormitory become friends and all prefer the same pizza place.
4. A smoker quits after befriending non-smokers.

For each: what data (cross-sectional vs. longitudinal, observational vs. experimental) is needed to identify the mechanism?

Solution:

1. Jazz → friends → classical: This contains **selection** and possibly **socialization**. The shared jazz taste existed *before* friendship and may explain why the friendship formed. The later shared interest in classical music could be peer influence, but confirming that requires longitudinal data on preferences before and after friendship formation.

2. Python users, day-1 friendship: Primarily **confounding**. Both speak Python because of the workplace; friendship arose from co-presence. Comparing pre-hire language use would help isolate the confound.

3. Dormitory pizza preference: Mostly **socialization** with **confounding**. Proximity (dorm) is the confound; shared experiences create the convergent preference. Comparing same-floor vs. different-floor pairs partially isolates proximity from preference convergence.

4. Smoker quits: Classic **socialization** (peer influence), but **selection** is plausible — the smoker may already have wanted to quit and selected a friendlier environment. Random assignment of smokers to friend groups would identify the mechanism, but raises ethical issues.

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Exercise 6.C.2: Designing a Study

You want to know whether students in the same study group become more similar in performance (socialization) or whether similar students form study groups together (selection).

1. What cross-sectional data would you collect?
2. What longitudinal data is needed to separate the mechanisms?
3. What would an ideal experimental design look like?
4. What would you observe if **both** mechanisms operate?

Solution:

1. Cross-sectional: Group membership and current GPA. You'd find that group members have more similar GPAs than non-members — but cannot distinguish selection from socialization.

2. Longitudinal: Measure GPA *before* groups form, record group formation, measure GPA *after*. Selection: groups whose pre-formation similarity is higher form more often. Socialization: GPA similarity increases more within groups than between random pairs.

3. Experimental: Randomly assign students to groups, eliminating selection. Any convergence is socialization. Ethical concerns: forced groupings; randomized encouragement designs are a softer alternative.

4. Both mechanisms: Pre-formation, groups already show above-average similarity (selection). Post-formation, similarity increases further (socialization). Initially-similar groups may converge *more*, indicating selection-by-socialization interaction.

6.D Spatial Segregation

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Exercise 6.D.1: Schelling Model Analysis

In Schelling's model, agents on a grid prefer at least k of 8 neighbours to be the same type. Dissatisfied agents move.

1. If $k = 1$, do you expect strong segregation? Why?
2. If $k = 5$ (majority), what outcome do you predict?
3. Why is it surprising that mild preferences ($k = 3$ or $k = 4$) can still produce strong global segregation?
4. How does this connect to homophily? What is the "network" in Schelling's model?

Solution:

1. $k = 1$: Almost no agents are dissatisfied. **Minimal segregation** — close to the random initial state.

2. $k = 5$: Most agents in mixed neighbourhoods are dissatisfied. Mass movement produces **near-complete segregation** into large monochromatic patches.

3. The surprise at $k = 3$ or $k = 4$: A 37–50% threshold *sounds* tolerant. Yet small same-type clusters become satisfied, attracting more similar agents and growing. Individual tolerance does not prevent collective segregation. **You cannot read individual preferences from aggregate outcomes.**

4. Connection: The Schelling grid *is* a network — cells are nodes, neighbouring cells are edges. Moving rewires ties. The preference rule is exactly homophily. Strong segregation emerges from local homophily at scale ([?EasleyKleinberg2010]).

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Exercise 6.D.2: Schelling Simulation

Implement a **1D Schelling model**: agents sit on a ring with local neighbourhood radius 4, so each occupied position checks up to 8 nearby positions. Each agent needs at least `threshold` fraction of occupied neighbours to be the same type. Unhappy agents move to random empty positions.

1. Implement and run for 100 steps.
2. Visualise initial and final states.
3. Vary `threshold` from 0.1 to 0.9 and plot average run-length at convergence.
4. At what threshold does strong segregation emerge?

Solution:

```
import random
import matplotlib.pyplot as plt

def schelling_1d(n=120, fraction_A=0.45, empty_fraction=0.10,
                 threshold=0.4, radius=4, steps=200, seed=None):
    rng = random.Random(seed)
    n_empty = round(n * empty_fraction)
    n_agents = n - n_empty
    n_A = round(n_agents * fraction_A)
    state = ["A"] * n_A + ["B"] * (n_agents - n_A) + [None] * n_empty
    rng.shuffle(state)
    initial = state.copy()

    for _ in range(steps):
        unhappy = []
        empty_positions = [i for i, value in enumerate(state) if value is None]

        for i in range(n):
            if state[i] is None:
```

With radius 4, strong segregation usually starts around thresholds of 0.4–0.5: contiguous same-type runs become much longer than in the random initial state. Very high thresholds such as 0.8 or 0.9 can be too demanding for this simple 1D model, so the system may keep moving rather than cleanly converging. The important lesson is still the Schelling one: moderate local similarity preferences can produce much stronger global segregation than the individual rule suggests.

Wrap-Up

- homophily is measurable; the E-I index is the basic diagnostic
- bipartite affiliations explain a lot of structure that pure friendship graphs hide
- selection, socialization, and confounding cannot be separated from cross-sectional data alone
- mild local preferences can still produce sharp global segregation